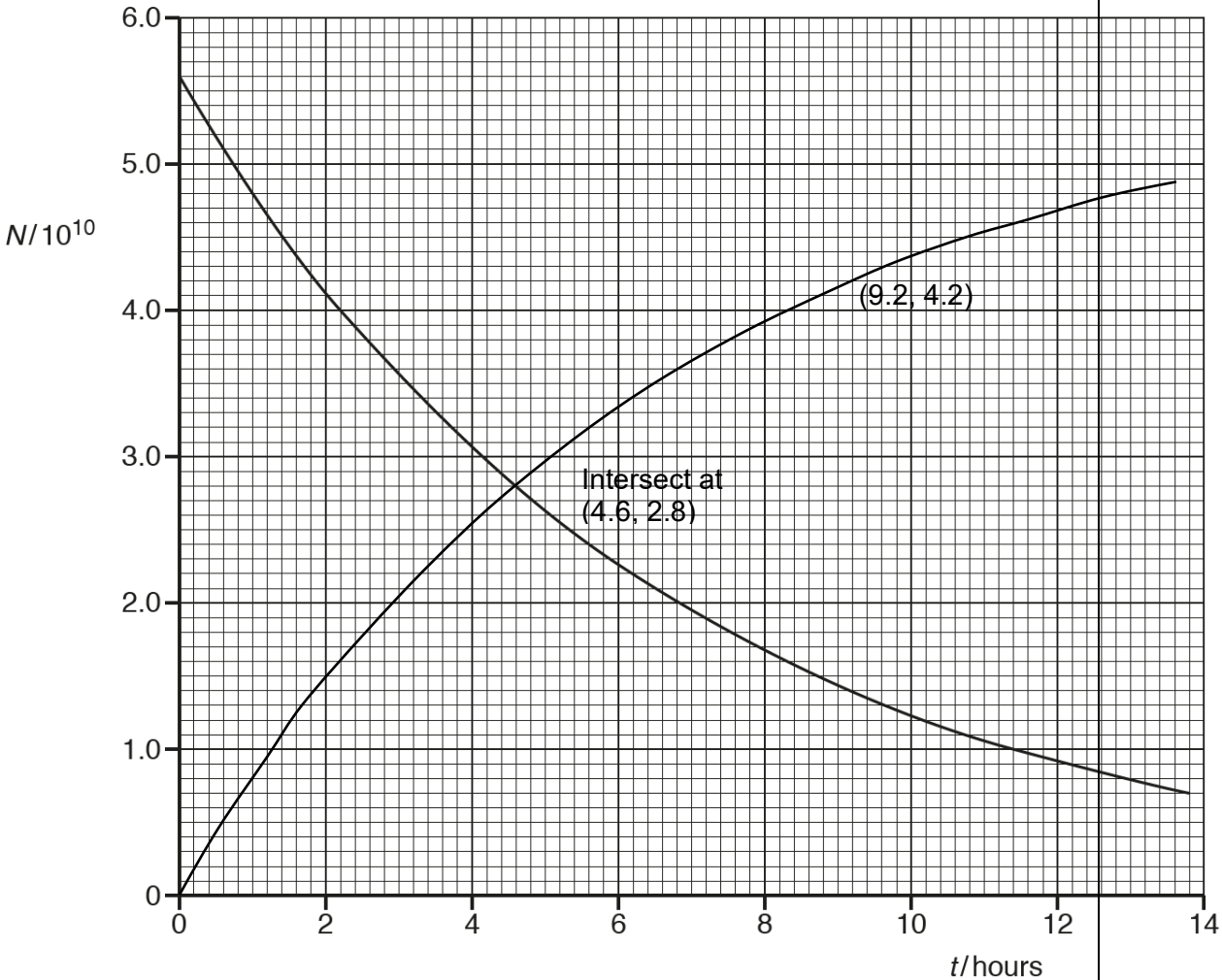
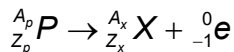
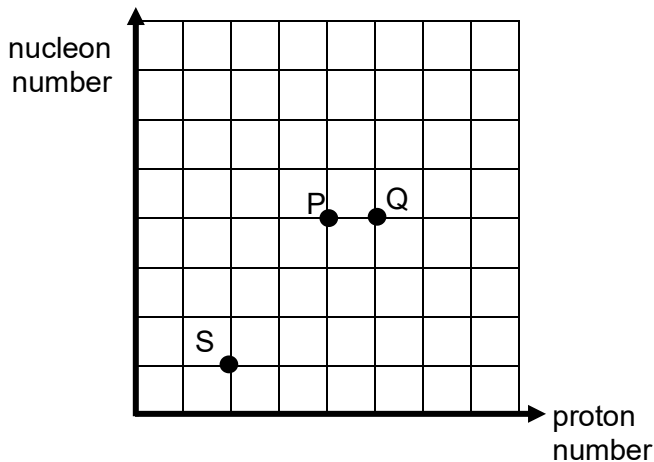


9a	Half-life is the <u>average</u> time taken for the <u>activity</u> of a particular radioactive <u>nuclide</u> to fall to half its initial value
9bi	Initial N = 5.6 At $\frac{1}{2}N$, t = 4.6 hours At $\frac{1}{4}N$, t = 9.2 hours Take average to find half-life $\text{Half-life} = \frac{4.6 + 4.6}{2} = 4.6$
9bii	$\frac{\text{number of X nuclei}}{\text{number of P nuclei}} = 6$ $\frac{\text{number of P nuclei after time } t}{\text{initial number of P nuclei}} = \frac{1}{7}$ $\left(\frac{1}{2}\right)^n = \frac{1}{7}$ $n = \frac{\lg\left(\frac{1}{7}\right)}{\lg\left(\frac{1}{2}\right)}$ $= 2.81$ Hence, t = n × half-life = 2.81 × 4.6 = 13 hours
9biii	(Negative) gradient of graph represents <u>activity / rate of</u> decay
9biv	Mtd 1: At t = 4.0, N = 3.05×10^{10} $A = \lambda N = \frac{\ln 2}{t_{\frac{1}{2}}} N$ $= \frac{\ln 2}{4.6 \times 3600} (3.05 \times 10^{10})$ $= 1.28 \times 10^6 \text{ Bq}$ OR Mtd 2: tangent drawn and gradient calculated $\text{Gradient} = \frac{(0.7 - 4.5) \times 10^{10}}{(9.4 - 0.8) \times 3600}$

	$= 1.23 \times 10^6 \text{ Bq}$
9bv	Correct trend Intersect at (4.6, 2.8) and Pass through (9.2, 4.2)  The graph shows two curves on a grid. The vertical axis is labeled $N/10^{10}$ and has major ticks every 1.0 unit from 0 to 6.0. The horizontal axis is labeled t/hours and has major ticks every 2 units from 0 to 14. One curve starts at approximately (0, 5.6) and decreases. The other curve starts at (0, 0) and increases. They intersect at the point (4.6, 2.8). A point on the increasing curve is labeled (9.2, 4.2).
9bvi	A neutrino is produced and has energy. Q has kinetic energy. <u>Mass-energy and momentum</u> must be conserved in the decay. The missing energy must have been carried away by other particles in the decay.

9ci
and
ii

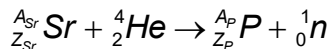


By conservation of mass:

$$A_p = A_x + 0 \Rightarrow A_p - A_x = 0$$

By conservation of charge:

$$Z_p = Z_x - 1 \Rightarrow Z_p - Z_x = -1$$



By conservation of mass:

$$A_{Sr} + 4 = A_p + 1 \Rightarrow A_{Sr} - A_p = -3$$

By conservation of charge:

$$Z_{Sr} + 2 = Z_p \Rightarrow Z_{Sr} - Z_p = -2$$

9cii

Total linear momentum of the system must be conserved throughout the process. Since initial total momentum of S and α is non-zero, final total momentum of them at closest approach must be non-zero. Hence, not possible for them to be at rest simultaneously.